

AI ACADEMIC MENTOR

Course	CSE 4104 — Software Development III
Team Name	CSE4104-7C-T05
Section	7C
Project Title	AI Academic Mentor — Personalized Learning & Study Assistant
Instructed By	Md. Riaz Mahmud, Assistant Professor, Dept. of CSE, NUBTK
GitHub Repository Link	cse4104-7c-t05-ai-academic-mentor/documentation at main · Priyo-1158/cse4104-7c-t05-ai-academic-mentor · GitHub
Document Type	Week 5 – Development Plan
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Team Information

SL No	Name	Student ID	Role	Contact Focus
1	Shadman Sadequeen Priyo	11230121158	Team Leader	Project coordination & GitHub
2	Musabbir Hossain Chayon	11230121168	Frontend Developer	React.js, Tailwind CSS, Figma
3	Sk. Nadirul Haque Nadir	11230121155	Backend Developer	Node.js, Express.js, MongoDB
4	Shafin Kabir	11230121175	AI Integration Lead	Gemini API, AI modules

Development Roadmap

The AI Academic Mentor project follows a 14-week development timeline structured across six phases. Each phase has defined deliverables, key tasks, assigned team members, and a completion status. The roadmap below serves as the primary reference during implementation and evaluation.

Week	Phase	Status	Deliverables	Key Tasks
Week 1–2	Planning	Completed	GitHub Repository, Project Proposal PDF	– Team formation and role assignment (Leader, Frontend, Backend, AI Integration Lead) – GitHub repo created with branch structure: main / development / frontend / backend / ai-integration – Technology stack finalized: React.js, Node.js, MongoDB, Gemini API – Week 2 Project Proposal document submitted
Week 3–5	Design	Completed	SRS Document, ER Diagram, Architecture Diagram, UI/UX Wireframes	– Week 3: Software Requirements Specification (SRS) + use case diagrams – Week 4: System architecture diagram, ER diagram, and REST API structure – Week 5: Figma wireframes for all 12 major screens – Week 5: UI Screenshots document and Development Planning document
Week 6–7	Development	Upcoming	Auth APIs, Database Schemas, All UI Pages Connected to Backend	– Week 6: JWT auth system, bcrypt password hashing, REST API endpoints – Week 6: MongoDB Atlas setup and database schemas for users, quizzes, study plans, progress – Week 7: Build all 12 UI pages in React.js + Tailwind CSS – Week 7: Connect all React pages to backend REST APIs and test data flow
Week 8–9	AI Integration	Upcoming	AI Chatbot, Quiz Generator, Note Summarizer, Study Planner, Dashboard	– Week 8: Integrate AI Chatbot (Gemini API) with conversation history per session – Week 8: Build AI Quiz Generator — MCQ + short-answer formats, score tracking in MongoDB – Week 8: Implement AI Note Summarizer — paste or upload notes, generate summaries – Week 9: Build AI-powered Personalized Study Planner

				with structured day-by-day plans – Week 9: Complete Progress Dashboard with charts, topic-wise analytics, and engagement metrics – Week 9: Polish UI, add in-app notification system and Admin Panel
Week 10–11	QA & Deploy	Upcoming	Testing Report, Live Deployment (Vercel + Render/Railway)	– Week 10: Write test cases; run unit, integration, and user acceptance testing – Week 10: Document all bugs and fixes in a testing report – Week 11: Deploy frontend to Vercel – Week 11: Deploy backend API to Render / Railway – Week 11: Verify live deployment and update README.md with deployment link
Week 12–14	Presentation	Upcoming	Final Report, User Manual, API Documentation, Live Demo	– Week 12: Write final project report, user manual, and API documentation – Week 13: Prepare presentation slides and rehearse live demonstration – Week 14: Final presentation, live demo, Q&A, and viva evaluation

Current Status Summary

Weeks 1–2 Planning	COMPLETED — GitHub repo created, Project Proposal submitted
Weeks 3–5 Design	COMPLETED — SRS, Architecture, ER Diagram, UI/UX Design documents done
Weeks 6–7 Development	UPCOMING — Backend API development + React.js frontend build
Weeks 8–9 AI Integration	UPCOMING — Gemini API integration, all 4 AI features
Weeks 10–11 QA & Deploy	UPCOMING — Testing, debugging, Vercel + Render/Railway deployment
Weeks 12–14 Presentation	UPCOMING — Final report, live demo, viva evaluation